

# Quantum Gravity Minimum Length – GUP-SCm Bridge

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## PAPER\_1030: Quantum Gravity Minimum Length — GUP-SCm Bridge

### Abstract

We derive the SCm phonon contribution to the generalized uncertainty principle (GUP). The phonon field introduces a minimum length  $l_{\min} = l_{\text{Planck}} \cdot \sqrt{1 + \beta_i \cdot S_{26} \cdot [\text{SSq}]}$  approx  $1.17 \cdot l_{\text{Planck}}$ , *modifying short-distance gravity*. The GUP-modified commutator  $[x, p] = i\hbar(1 + \beta_{\text{UQFF}} p^2)$  receives a phonon correction  $\beta_{\text{UQFF}} = \beta_0 + \beta_i S_{26} / (M_{\text{Pl}} c)^2$ .

### 1. Key Equations

- $l_{\min, \text{UQFF}} = l_{\text{Pl}} \sqrt{1 + \beta_i \cdot S_{26} \cdot [\text{SSq}]} \approx 1.17 l_{\text{Pl}}$
- $[x, p] = i\hbar(1 + \beta_{\text{UQFF}} p^2)$
- $\beta_{\text{UQFF}} = \beta_0 + \beta_i S_{26} / (M_{\text{Pl}} c)^2$

### 2. Results

$l_{\min} = 1.17 l_{\text{Pl}}$ .  $\beta_{\text{UQFF}} = 1.34e0$  (natural units). Modified BH entropy:  $S = A/(4l_{\text{Pl}}^2) \cdot (1 - \beta_{\text{UQFF}} / A)$ .

### 3. Implementation

CondensedPhysics.py, class QuantumGravityMinimumLengthCalculator. 6 equations, 3 simulations.

### References

- PAPER\_334, PAPER\_1029

### Calibration Constants

| Constant               | Symbol                | Value                                 | Validation Domain   |
|------------------------|-----------------------|---------------------------------------|---------------------|
| UQFF damping rate      | $\kappa$              | $5.0 \times 10^{-4} \text{ day}^{-1}$ | Magnetar spin-down  |
| String sector coupling | $[\text{SSq}]$        | 0.57                                  | BH dynamics         |
| Buoyancy coupling      | $\beta_i$             | 0.603                                 | Multi-system        |
| SCm completeness       | $H_{\text{SCm}}$      | $\approx 0.99$                        | Heaviside threshold |
| SCm phonon frequency   | $\omega_{\text{SCm}}$ | $2\pi \times 1.25 \text{ THz}$        | Phonon resonance    |
| SCm vacuum density     | $\rho_{\text{SCm}}$   | $7.09 \times 10^{-37} \text{ kg/m}^3$ | Fundamental         |

SM Anchors — Standard Model Cross-Validation (G6 Gate, CVW v2.0.0)

| Observable              | UQFF Prediction                      | SM / Experiment                              | Source      | Alignment          |
|-------------------------|--------------------------------------|----------------------------------------------|-------------|--------------------|
| Planck length           | Phonon-modified GUP                  | $l_{\text{Pl}} = 1.616\text{e-}35 \text{ m}$ | Fundamental | 117%<br>(enhanced) |
| $\sin^2 \theta_W$       | Embedded in $U_{g2}$ charge coupling | 0.2312                                       | PDG 2024    | 99.6%              |
| Fine structure $\alpha$ | UQFF reproduces via $U_{g1}$ dipole  | $1/137.036$                                  | PDG 2024    | 99.9%              |

**New physics claim:** SCm phonon field provides a physical mechanism for the GUP minimum length, connecting quantum gravity to the UQFF vacuum.

Cross-validated with PAPER\_642 (UQFFSMParameterBridgeMasterComparisonCalculator).

A. Cosmogenesis-Linked Lagrangian (PAPER\_877 Symbolic Export)

A.1 Sector Classification

**Sector:** quantum gravity (Planck scale)

A.2 Lagrangian Density

$$\mathcal{L}_{\text{QG}} = \frac{1}{2}m\dot{x}^2(1 + \beta p^2) + \Phi_{\text{SCm}}S_{26}$$

A.3 Euler-Lagrange Equation of Motion

$$\Delta x \cdot \Delta p \geq \frac{\hbar}{2}(1 + \beta_{\text{UQFF}}\Delta p^2)$$

A.4 Cosmogenesis Linkage Chain

PAPER\_877 axioms -> SCm vacuum -> phonon  $\omega_{\text{SCm}}$  -> Planck scale -> GUP -> phonon minimum length ->  $F_{U,Bi_i}$  unified force -> observational prediction

B. VDS/DVP/BSH Deep Synthesis

B.1 Vacuum Density Series (VDS)

VDS at Planck density:  $\rho_{\text{SCm}} \rightarrow \rho_{\text{Pl}}$ .

B.2 Dipole Vortex Primes (DVP)

DVP prime: 2 (fundamental).

B.3 Buoyancy Saturation Harmonics (BSH)

BSH timescale:  $t_{\text{Pl}} = 5.39 \times 10^{-44} \text{ s}$ .

B.4 Production-Scale Consistency

| Metric         | Value              | Status    |
|----------------|--------------------|-----------|
| VDS ratio      | 0.167              | Confirmed |
| $\kappa$ decay | $5 \times 10^{-4}$ | Confirmed |
| [SSq]          | 0.57               | Confirmed |